

1. Use a complete truth table to determine whether the following compound statement is tautologous, self-contradictory or contingent. (8marks)
 $((P \supset \sim Q) \vee S) \cdot P$
2. Use a complete truth table to determine whether the following statements are logically equivalent, contradictory, consistent, or inconsistent. (7 marks)
 $\sim(P \supset Q)$ and $\sim Q \cdot P$
3. Use indirect truth tables to determine whether the following arguments are valid. (10marks)
 $(S \vee N) \supset L / M \cdot \sim L // \sim N \cdot S$

This image shows a full page of blank graph paper. The grid consists of small, identical squares formed by thin black lines. There are 20 columns and 20 rows of squares, creating a total of 400 square units. The margins are consistent on all sides, and there are no markings or text on the paper.

